

QNPSA produces irreproducible results

<https://github.com/Qiskit/qiskit/issues/7069>

ROW 67

QNPSA produces irreproducible results #7069

[New issue](#)Closed#7071

Zoufalc opened on Sep 28, 2021 · edited by Zoufalc

Edits ...

Information

- Qiskit Terra version: qiskit-terra 0.17.0
- Python version: Python 3.7.3
- Operating system: MacOS Big Sur 11.6

What is the current behavior?

Executing a QNPSA optimization with the same random seed gives different results. We checked that the same optimization gives reproducible results with other optimizers such as ADAM.

Steps to reproduce the problem

Run an arbitrary optimization, e.g., VQE with QNPSA, and compare the results.

What is the expected behavior?

The same results for the same seed.

Suggested solutions



Zoufalc added bug on Sep 28, 2021

Assignees

No one assigned

Labels

bug

Type

No type

Projects

No projects

Milestone

0.19

Closed on Dec 6, 2021, 100% complete

Relationships

None yet

Development

Add note on reproducibility to QNPSA
Qiskit/qiskit

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Participants



jakelishman on Sep 28, 2021

Member ...

Could you post a minimal reproducer?

The [documentation for SPSA](#) (from which QNPSA derives) says

This component has some function that is normally random. If you want to reproduce behavior then you should set the random number generator seed in the algorithm_globals (`qiskit.utils.algorithm_globals.random_seed = seed`).

Could that have anything to do with it?



javabster on Sep 28, 2021

Contributor ...

When I run the following VQE with QNPSA I get consistent results, as @jakelishman said if you're not setting your seed using the `algorithm_globals.random_seed` that could be the cause.

```
seed = 50
algorithm_globals.random_seed = seed
qi = QuantumInstance(Aer.get_backend('statevector_simulator'), seed_transpiler=seed, seed_simulator=seed)

ansatz = PauliTwoDesign(2, reps=1, seed=seed)

fidelity = QNPSA.get_fidelity(ansatz)
my_opt = QNPSA(fidelity, maxiter=1000)

vqe = VQE(ansatz, optimizer=my_opt, quantum_instance=qi)
result = vqe.compute_minimum_eigenvalue(operator=H2_op)
print(result)
```



```
fidelity = QNSPSA.get_fidelity(ansatz)
my_opt = QNSPSA(fidelity, maxiter=1000)

vqe = VQE(ansatz, optimizer=my_opt, quantum_instance=qi)
result = vqe.compute_minimum_eigenvalue(operator=H2_op)
print(result)
optimizer_evals = result.optimizer_evals
```

2 example results:

```
{ 'aux_operator_eigenvalues': None,
  'cost_function_evals': 7001,
  'eigenstate': array([ 8.53553391e-01+1.14967765e-08j, -1.05132805e-09-3.53553391e-01j,
    -1.05132866e-09+3.53553391e-01j, -1.46446609e-01+1.97253476e-09j]),
  'eigenvalue': (-1.1484788977930933+0j),
  'optimal_parameters': { ParameterVectorElement(0[1]): 1.5645826144607016,
    ParameterVectorElement(0[2]): 25.96335152999994,
    ParameterVectorElement(0[3]): 6.2893990090180765,
    ParameterVectorElement(0[0]): -21.250962566058167},
  'optimal_point': array([-21.25096257,  1.56458261,  25.96335153,  6.28939901]),
  'optimal_value': -1.1484788977930933,
  'optimizer_evals': None,
  'optimizer_time': 35.11342716217041}
```

```
{ 'aux_operator_eigenvalues': None,
  'cost_function_evals': 7001,
  'eigenstate': array([ 8.53553391e-01+1.14967765e-08j, -1.05132805e-09-3.53553391e-01j,
    -1.05132866e-09+3.53553391e-01j, -1.46446609e-01+1.97253476e-09j]),
  'eigenvalue': (-1.1484788977930933+0j),
  'optimal_parameters': { ParameterVectorElement(0[1]): 1.5645826144607016,
    ParameterVectorElement(0[2]): 25.96335152999994,
    ParameterVectorElement(0[3]): 6.2893990090180765,
    ParameterVectorElement(0[0]): -21.250962566058167},
  'optimal_point': array([-21.25096257,  1.56458261,  25.96335153,  6.28939901]),
  'optimal_value': -1.1484788977930933,
  'optimizer_evals': None,
  'optimizer_time': 35.11342716217041}
```



jakelishman on Sep 28, 2021

Member ...

We should make sure that the documentation for the `QNSPSA` class also includes the same note about setting the global seed, since users probably won't see it with it just being on a parent class. Hopefully this is the issue here.



2



Zoufalc on Sep 28, 2021

Contributor Author ...

Great! That resolved the problem. I was not aware of the global setting. Thanks for pointing this out.



Zoufalc closed this as **completed** on Sep 28, 2021



jakelishman on Sep 28, 2021

Member ...

Great to hear! I'll just leave this open for now as a reminder to ourselves to update the documentation in the class.



1



jakelishman reopened this on Sep 28, 2021



jakelishman mentioned this on Sep 28, 2021

[Add note on reproducibility to QNSPSA #7071](#)

Add note on reproducibility to QNSPSA #7071

<> Code

Merged mergify merged 2 commits into `Qiskit:main` from `jakelishman:fix/documentation-qnspsa-seed` on Sep 28, 2021

Conversation 0 Commits 2 Checks 0 Files changed 1 +6 -0



jakelishman commented on Sep 28, 2021

Member

Summary

This class inherits from `SPSA`, which has a note about setting `qiskit.utils.algorithm_globals.random_seed` for reproducible behaviour, however this wasn't visible on the `QNSPSA` documentation page. This commit just copies the same note up.

Details and comments

Fix #7069



Add note on reproducibility to QNSPSA **Verified** 66ffe8

jakelishman requested review from manaelmarques, woodsp-ibm and a team as code owners 4 years ago

jakelishman added `Changelog: None` `stable backport potential` labels on Sep 28, 2021



1ucian0 approved these changes on Sep 28, 2021

View reviewed changes

1ucian0 added the `automerge` label on Sep 28, 2021

woodsp-ibm added the `mod: algorithms` label on Sep 28, 2021



woodsp-ibm approved these changes on Sep 28, 2021

View reviewed changes

Merge branch 'main' into fix/documentation-qnspsa-seed

Verified 5617cbf

jakelishman mentioned this pull request on Sep 28, 2021

Fix 1Q gate errors in `plot_error_map` #7037

Merged

mergify (bot) merged commit 364f121 into `Qiskit:main` on Sep 28, 2021

mergify (bot) pushed a commit that referenced this pull request on Sep 28, 2021

Add note on reproducibility to QNSPSA (#7071) **Unverified** cd966b5

mergify (bot) mentioned this pull request on Sep 28, 2021

Add note on reproducibility to QNSPSA (backport #7071) #7074

Merged

Reviewers

woodsp-ibm

1ucian0

manaelmarques

Assignees

No one assigned

Labels

`Changelog: None` `mod: algorithms` `stable backport potential`

Projects

None yet

Milestone

0.19

Development

Successfully merging this pull request may close these issues.

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4 participants

